

INCH-POUND

MS27653B  
W/AMENDMENT 2  
27 June 2025  
SUPERSEDING  
MS27653B  
W/AMENDMENT 1  
8 February 2008

# DETAIL SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, STRAIGHT,  
CRIMP TYPE, BAYONET COUPLING, SERIES I

Inactive for new design after 13 August 1974.  
For new design use MS27467.

This specification is approved for use by all Departments  
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein  
shall consist of this specification sheet and MIL-DTL-38999.

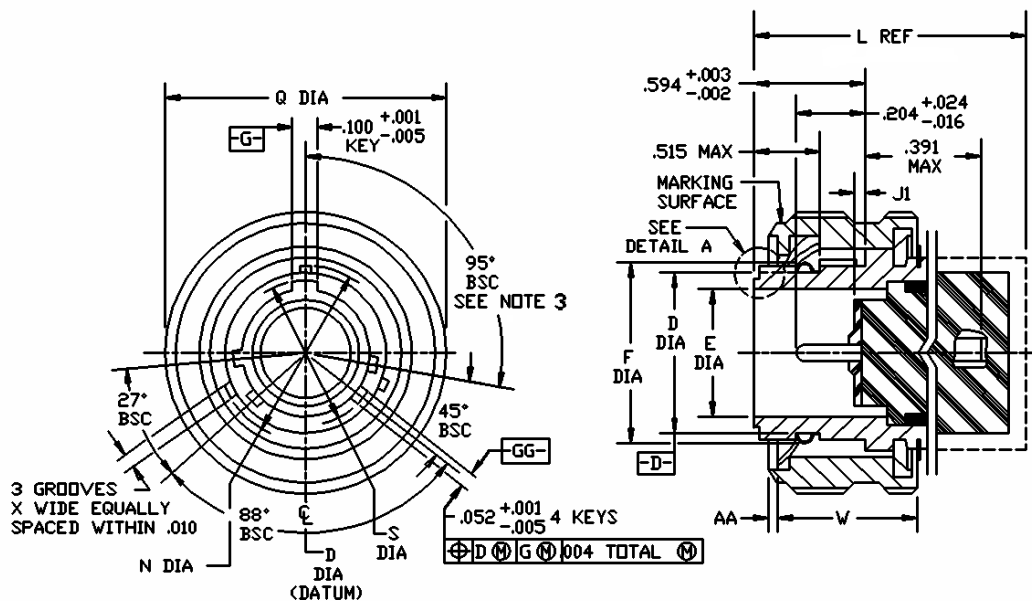


FIGURE 1. Plug, classes E and T.

MS27653B  
w/AMENDMENT 2

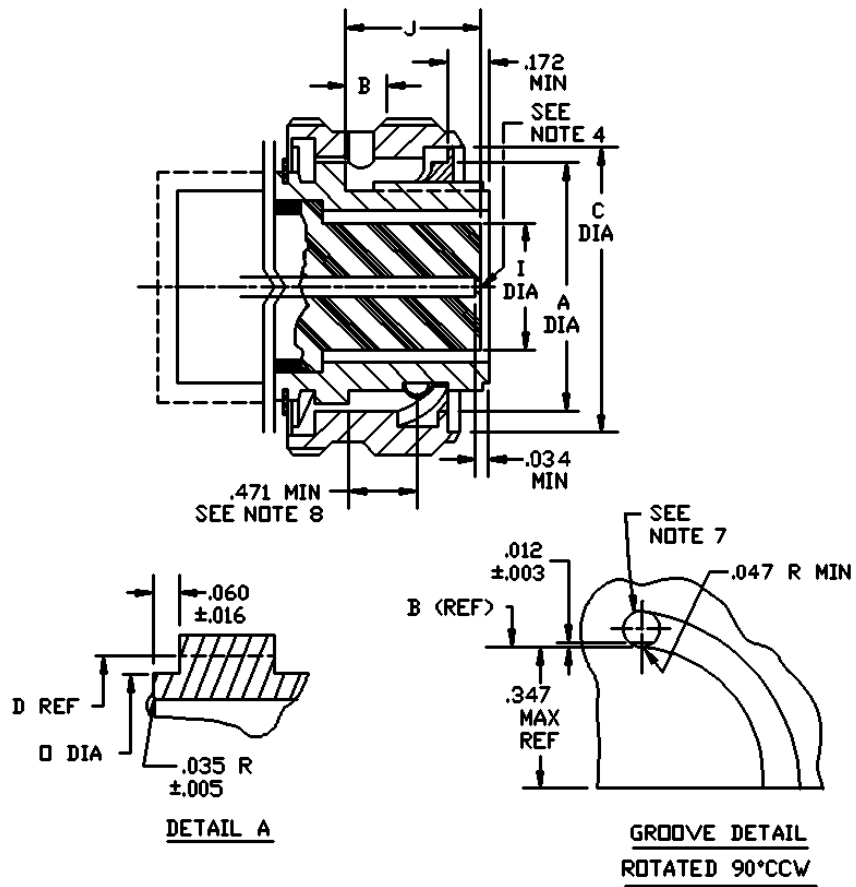


FIGURE 1. Plug, classes E and T – Continued.

MS27653B  
w/AMENDMENT 2

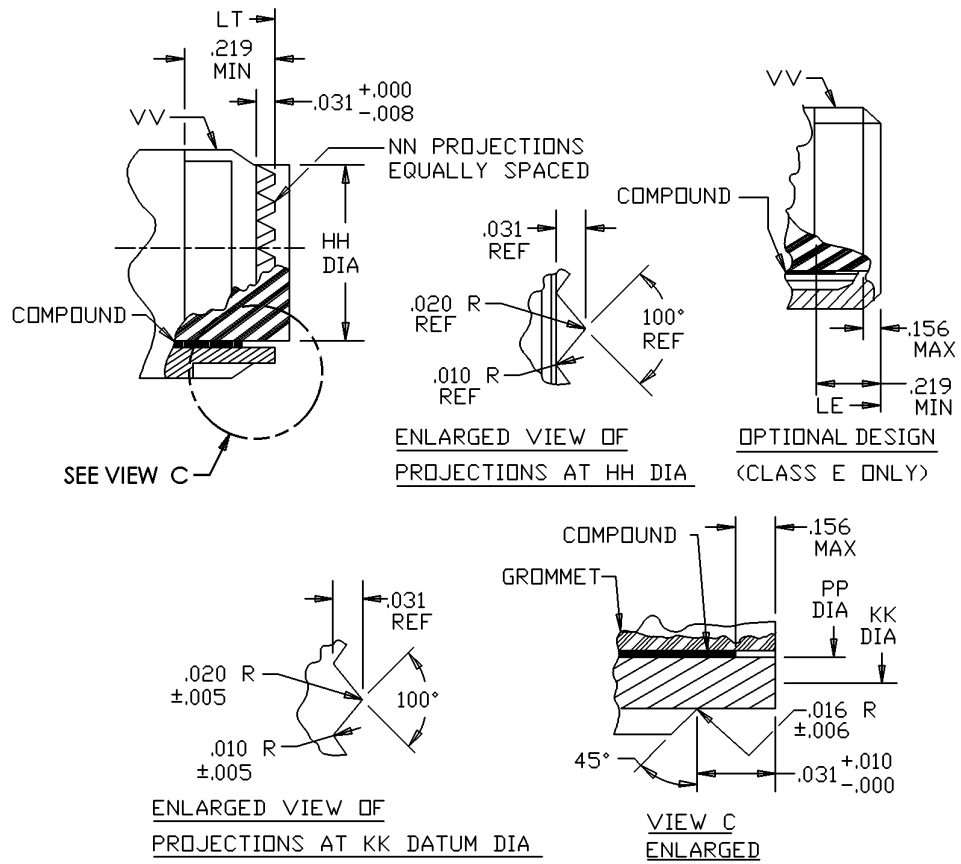


FIGURE 1. Plug, classes E and T – Continued.

MS27653B  
w/AMENDMENT 2

| Shell size | A dia<br>+.005<br>-.001 | B<br>±.020 | C dia<br>+.005<br>-.001 | D dia<br>+.001<br>-.005 | D <sup>1</sup> dia<br>min | E dia<br>±.003 | F dia<br>min | I dia<br>max | J<br>+.007<br>-.003<br>socket<br>insert | J <sup>1</sup><br>+.012<br>-.003<br>pin<br>insert |
|------------|-------------------------|------------|-------------------------|-------------------------|---------------------------|----------------|--------------|--------------|-----------------------------------------|---------------------------------------------------|
| 9          | .580                    | .113       | .684                    | .437                    | .417                      | .315           | .447         | .290         | .575                                    | .019                                              |
| 11         | .712                    |            | .832                    | .565                    | .545                      | .433           | .575         | .418         |                                         |                                                   |
| 13         | .861                    |            | .981                    | .677                    | .657                      | .547           | .692         | .532         |                                         |                                                   |
| 15         | .985                    |            | 1.106                   | .802                    | .782                      | .672           | .817         | .657         |                                         |                                                   |
| 17         | 1.110                   |            | 1.230                   | .927                    | .907                      | .797           | .942         | .782         |                                         |                                                   |
| 19         | 1.217                   |            | 1.347                   | 1.032                   | 1.012                     | .886           | 1.047        | .871         |                                         |                                                   |
| 21         | 1.342                   |            | 1.462                   | 1.157                   | 1.137                     | 1.011          | 1.172        | .996         |                                         |                                                   |
| 23         | 1.467                   |            | 1.587                   | 1.282                   | 1.262                     | 1.136          | 1.297        | 1.121        |                                         |                                                   |
| 25         | 1.592                   | .152       | 1.712                   | 1.407                   | 1.387                     | 1.260          | 1.442        | 1.246        |                                         |                                                   |

| Shell size | N dia<br>+.005<br>-.006 | Q dia<br>max | S dia<br>+.001<br>-.010 | W<br>+.011<br>-.006 | X<br>+.016<br>-.001 | AA<br>+.005<br>-.006 | For insert arrangement<br>see<br>MIL-STD-1560 |
|------------|-------------------------|--------------|-------------------------|---------------------|---------------------|----------------------|-----------------------------------------------|
| 9          | .674                    | .859         | .482                    | .693                | .090                | .068                 |                                               |
| 11         | .822                    | .984         | .610                    | .682                |                     |                      |                                               |
| 13         | .972                    | 1.156        | .759                    |                     |                     |                      |                                               |
| 15         | 1.097                   | 1.281        | .884                    | .693                |                     |                      |                                               |
| 17         | 1.222                   | 1.406        | 1.009                   |                     |                     |                      |                                               |
| 19         | 1.329                   | 1.516        | 1.114                   |                     |                     |                      |                                               |
| 21         | 1.454                   | 1.641        | 1.239                   |                     |                     |                      |                                               |
| 23         | 1.579                   | 1.766        | 1.364                   |                     |                     |                      |                                               |
| 25         | 1.704                   | 1.891        | 1.489                   | .794                | .137                | .047                 |                                               |

FIGURE 1. Plug, classes E and T – Continued.

MS27653B  
w/AMENDMENT 2

| Shell size | HH dia<br>+.000<br>-.005 | KK DATUM dia | NN | PP dia<br>± .003 | LE max overall length | LT max overall length | VV Thread class 2A (plated) |
|------------|--------------------------|--------------|----|------------------|-----------------------|-----------------------|-----------------------------|
| 9          | .387                     | .366         | 12 | .344             | 1.234                 | 1.234                 | 7/16-28                     |
| 11         | .515                     | .487         | 16 | .472             |                       |                       | 9/16-24                     |
| 13         | .628                     | .609         | 20 | .586             |                       |                       | 11/16-24                    |
| 15         | .754                     | .731         | 24 | .711             |                       |                       | 13/16-20                    |
| 17         | .879                     | .853         | 28 | .836             |                       |                       | 15/16-20                    |
| 19         | .985                     | .974         | 32 | .942             |                       |                       | 1 1/16-18                   |
| 21         | 1.110                    | 1.097        | 36 | 1.067            |                       |                       | 1 3/16-18                   |
| 23         | 1.235                    | 1.218        | 40 | 1.192            |                       |                       | 1 5/16-18                   |
| 25         | 1.360                    | 1.340        | 44 | 1.317            |                       |                       | 1 7/16-18                   |

| Inches | mm   | Inches | mm   | Inches | mm    | Inches | mm     |
|--------|------|--------|------|--------|-------|--------|--------|
| .001   | .025 | .010   | .254 | .035   | .889  | .204   | 5.182  |
| .002   | .508 | .012   | .305 | .047   | 1.194 | .219   | 5.563  |
| .003   | .077 | .016   | .406 | .052   | 1.321 | .347   | 8.814  |
| .004   | .102 | .020   | .508 | .060   | 1.524 | .391   | 9.931  |
| .005   | .013 | .024   | .609 | .100   | 2.540 | .471   | 11.963 |
| .006   | .152 | .031   | .787 | .156   | 3.962 | .594   | 15.088 |
| .008   | .203 | .034   | .864 | .172   | 4.369 |        |        |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Normal keyway position. For other keyway positions, MIL-DTL-38999.
4. Insert front surface shall be flat within .005 Total Indicator Reading (TIR).
5. Diameters A and N shall be concentric within .008 TIR.
6. Diameters D and I shall be concentric within .015 TIR at Maximum Material Condition (MMC).
7. 3 holes (design optional) equally spaced within .010 shall be provided for visual inspection of lock when mated with receptacle.
8. The point at which the gage pin, having the same basic diameter as the mating contact and a square face, touches socket contact spring.

FIGURE 1. Plug, classes E and T – Continued.

MS27653B  
w/AMENDMENT 2

REQUIREMENTS:

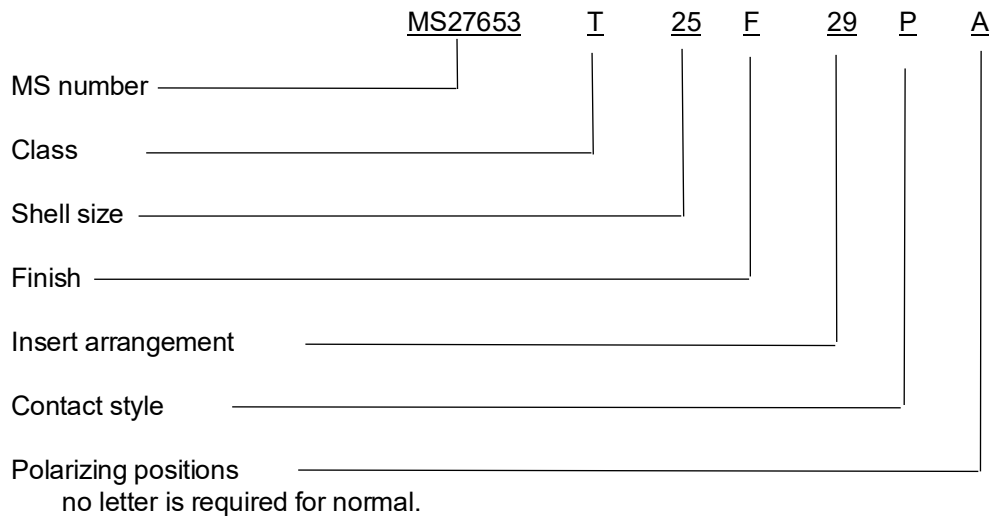
Dimensions and configuration: See figure 1.

Interface dimensions shall conform to MIL-DTL-38999.

Mating connectors: MS27466, MS27468, MS27469, MS27470, MS27471, MS27496, MS27505, MS27515, MS27652, MS27654, MS27656 and MS27662.

Insert arrangements: See MIL-STD-1560.

Part or Identifying Number (PIN) example:



Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MS27466  
MS27467  
MS27468  
MS27469  
MS27470  
MS27471  
MS27496  
MS27505  
MS27515  
MS27652  
MS27654  
MS27656  
MS27662  
MIL-STD-1560

MS27653B  
w/AMENDMENT 2

CONCLUDING MATERIAL

Custodians:

Army - CR  
Navy - AS  
Air Force - 85  
DLA - CC

Preparing activity:  
DLA - CC

(Project 5935-2025-055)

Review activities:

Army - AR, MI  
Navy - EC, MC, OS  
Air Force - 19

NOTE: The activities listed above were interested in this document as of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at <https://assist.dla.mil>.